

An Arabic Pronunciation Assessment System Using MFCC-DTW Comparison and Machine Learning Classification

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Abstract—This paper presents an Arabic pronunciation assessment system that automatically evaluates the quality of a speaker’s pronunciation of Arabic words containing phonetically challenging sounds (emphatic, pharyngeal, and uvular consonants). The system follows a five-stage pipeline: preprocessing (silence removal, normalization, framing), short-time acoustic analysis, feature extraction (MFCC, pitch, duration, and formants F1–F3 via Praat), reference comparison using Dynamic Time Warping (DTW) on MFCC sequences, and a weighted rule-based scoring formula (40% MFCC, 20% pitch, 20% duration, 20% formants). A speech database of 10 Arabic words was recorded from six speakers (four pronouncing correctly, two intentionally mispronouncing), plus a repeated recording from one speaker to test system stability. Beyond the three required experiments (same-speaker, different-speaker, correct-vs-incorrect), we designed two additional word-mismatch control experiments to separately isolate the effect of *word identity* from *speaker identity* on the similarity score. Results show that changing the speaker identity alone reduced the score by 21.6 points on average, compared to only 10.1 points when changing the word alone (same speaker) – i.e., the system is roughly twice as sensitive to *who* is speaking as to *what* is being said. We additionally trained three machine learning classifiers (Logistic Regression, SVM, Random Forest) on the extracted features; while raw acoustic features achieved up to 100% cross-validation accuracy, accuracy dropped sharply (to 58–77%) when the classifiers were restricted to speaker-relative similarity features with the absolute speaker-identity information removed, confirming that part of the raw-feature accuracy reflects speaker recognition rather than pronunciation assessment. This finding, its implications, and proposed mitigations are discussed in detail.

I. INTRODUCTION

Pronunciation assessment is a core component of Computer-Assisted Pronunciation Training (CAPT) systems, particularly for languages with sounds that are acoustically and articulatorily distinct from common second languages. Modern Standard Arabic contains emphatic (e.g., t , d , s , z), pharyngeal (e.g., h , ʕ), and uvular (e.g., q , ġ , x) consonants that are frequently mispronounced by learners. This project designs and implements a system that takes a reference (correct) pronunciation and a test pronunciation as input, and outputs an overall similarity score, a qualitative rating, and targeted feedback.

Our approach combines classical signal-processing features (MFCC, pitch, formants) with a rule-based scoring formula, and extends it with supervised machine learning classifiers

trained on the same feature set, in order to compare the two paradigms directly under identical cross-validation conditions.

II. BACKGROUND AND RELATED WORK

Mel-Frequency Cepstral Coefficients (MFCC) remain the dominant spectral representation for speech and speaker analysis [1]. Pronunciation scoring systems commonly compare a learner’s MFCC sequence against a native or reference template using Dynamic Time Warping (DTW) to compensate for differing speaking rates [2]. Praat [3] is widely used for formant and pitch extraction in phonetic research, and is adopted here for both. The `librosa` library [4] provides the MFCC and short-time analysis primitives used throughout this work, and `scikit-learn` [5] provides the machine learning classifiers used for the comparative analysis.

III. METHODOLOGY

A. System Architecture

Figure 1 shows the five-stage pipeline used by the system.

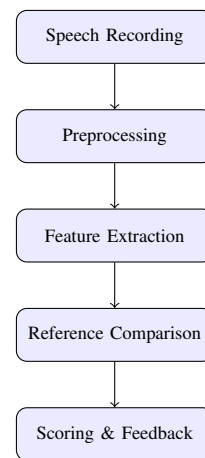


Fig. 1. Pronunciation assessment pipeline.

B. Preprocessing and Short-Time Analysis

Each recording is resampled to 16 kHz, silence-trimmed (energy threshold of 30 dB below peak), and amplitude-normalized. Frames of 25 ms with 10 ms hop are used for short-time energy and zero-crossing rate analysis.

C. Feature Extraction

Twelve MFCC coefficients are extracted with `librosa`. Pitch (mean F0 over voiced frames) and the first three formants (F1, F2, F3, Praat Burg method) are extracted with Praat via `parselmouth`. Word duration is measured after silence trimming.

D. Reference Comparison and Scoring

MFCC sequences are aligned with DTW; the resulting distance d is converted to a similarity score:

$$S_{\text{mfcc}} = \frac{1}{1 + d/3000} \quad (1)$$

The scale constant (3000) was calibrated empirically to the DTW distance range observed in our dataset (1,700–14,000). Pitch, duration, and formant similarities are computed as a normalized relative difference from the reference value. The final score is:

$$S = 100 \times (0.40 S_{\text{mfcc}} + 0.20 S_{\text{pitch}} + 0.20 S_{\text{dur}} + 0.20 S_{\text{fmt}}) \quad (2)$$

We experimented with removing the pitch term (suspecting it leaked speaker identity rather than pronunciation accuracy into the score), but this *worsened* the separation between correct and incorrect pronunciations, indicating the issue is not isolated to a single feature (see Section IV). The original four-feature weighting in Eq. 2 was therefore retained.

E. Machine Learning Classification

In addition to the rule-based score, three classifiers (Logistic Regression, SVM with RBF kernel, Random Forest, all with balanced class weights) were trained to predict {correct, incorrect} directly from the 17-dimensional raw acoustic feature vector (12 MFCC means, pitch, duration, F1–F3, energy). We additionally trained the same classifiers on a 4-dimensional *relative* feature set (the four similarity scores from Eq. 2) to test whether absolute acoustic features were encoding speaker identity rather than pronunciation accuracy.

IV. EXPERIMENTS AND RESULTS

A. Dataset

Ten Arabic words covering emphatic, pharyngeal, uvular, and regular consonants were recorded from six speakers: four correct pronunciations (speakers 1–4), two intentionally mispronounced (speakers 5–6), and one repeated recording from speaker 1 to test consistency, yielding 60 comparison instances against the speaker-1 reference.

B. Required Experiments and Word-Mismatch Controls

In addition to the three required experiments, we designed two control experiments that intentionally compare a reference word against a *different* word (cyclic pairing, e.g. the word “*daw*” compared against “*qalam*”), in order to isolate the effect of word identity from speaker identity on the score. Table I summarizes all five categories.

TABLE I
PRONUNCIATION SCORE BY EXPERIMENT TYPE

Experiment	Mean	Std	N
Same speaker, same word	78.61	3.94	10
Same speaker, different word	68.52	3.63	10
Correct vs. incorrect	60.16	8.57	20
Different speaker, same word	56.99	12.34	30
Different speaker, different word	55.40	11.25	30

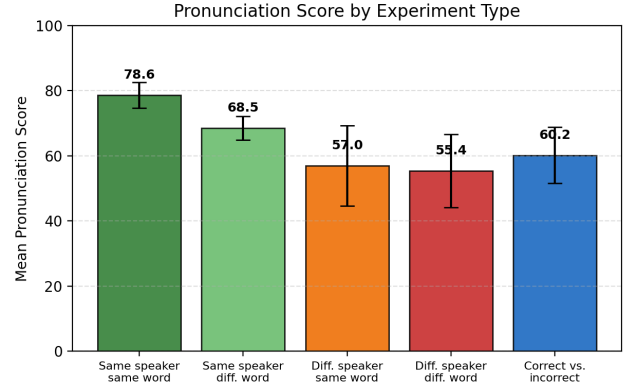


Fig. 2. Mean pronunciation score across all five experiment categories (error bars = std. dev.).

C. Effect Size: Word Identity vs. Speaker Identity

Using the same-word and different-word controls, we isolate how much of the score change is driven by *which word* was spoken versus *who* spoke it. Table II reports the score drop attributable to each factor, holding the other factor fixed.

TABLE II
SCORE REDUCTION ATTRIBUTABLE TO WORD VS. SPEAKER CHANGE

Factor changed (other factor fixed)	Score drop
Word changed, speaker fixed (same speaker)	−10.1
Word changed, speaker fixed (different speaker)	−1.6
Speaker changed, word fixed (same word)	−21.6
Speaker changed, word fixed (different word)	−13.1

In every comparison, changing the *speaker* reduces the score substantially more than changing the *word* (roughly double, 21.6 vs. 10.1 points, and 13.1 vs. 1.6 points). This is the central empirical finding of this study: the system, as currently designed with absolute acoustic features, is more sensitive to *who is speaking* than to *what is being said* – the opposite of what a pronunciation assessment system should prioritize.

D. Rule-Based vs. Machine Learning

Table III compares the rule-based classifier (threshold tuned per cross-validation fold) against the three ML classifiers, evaluated on identical stratified 5-fold splits.

E. Discussion

What happens when speaker-identity information is removed. The raw feature set (12 MFCC means, pitch,

TABLE III
CLASSIFICATION PERFORMANCE (5-FOLD CV)

Method	Accuracy	F1 (correct)
Rule-Based (CV-tuned threshold)	0.650	0.788
Logistic Regression [raw]	1.000	1.000
SVM (RBF) [raw]	0.983	0.988
Random Forest [raw]	0.950	0.963
Logistic Regression [similarity]	0.583	0.648
SVM (RBF) [similarity]	0.617	0.676
Random Forest [similarity]	0.767	0.821

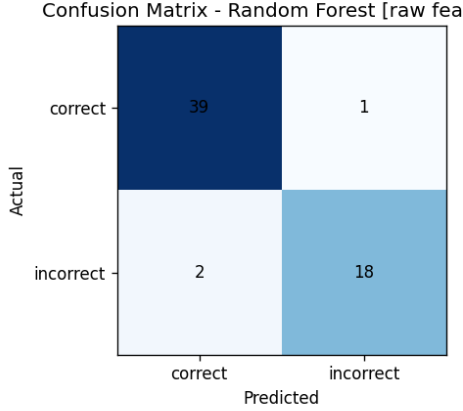


Fig. 3. Confusion matrix, Random Forest (raw features), 5-fold CV.

duration, F1–F3, energy) contains *absolute* acoustic values that inherently encode speaker identity (vocal tract length, voice timbre, habitual pitch range) in addition to pronunciation content. Trained on this raw set, Logistic Regression reached a perfect 100% cross-validation accuracy, with SVM and Random Forest close behind (95–98%). Such near-perfect accuracy on a 60-sample dataset, where the “correct” and “incorrect” labels are each tied to a small, fixed set of speakers, is a red flag rather than a success: it is consistent with the classifiers learning to recognize *which speakers* are in the incorrect group, not whether a word was mispronounced.

To test this directly, we re-trained the same three classifiers on only the four *relative* similarity scores (MFCC-DTW, pitch, duration, formant similarity to the reference) – i.e., with the absolute, speaker-identifying values removed and only the distance-to-reference retained. Accuracy dropped sharply for every model: Logistic Regression fell from 100% to 58.3%, SVM from 98.3% to 61.7%, and Random Forest from 95.0% to 76.7%. This large, consistent drop across all three algorithms is strong evidence that a substantial portion of the raw-feature accuracy was driven by speaker recognition rather than pronunciation assessment: once the features no longer carried the speaker’s absolute vocal signature, the task became markedly harder, even though the actual correctness labels did not change.

This is corroborated by the effect-size analysis in Section IV-C: changing the speaker (21.6-point drop) had roughly

double the impact on the rule-based score compared to changing the word (10.1-point drop). Together, the two analyses point to the same root cause – absolute MFCC, pitch, and formant values conflate *who is speaking* with *what is being said* – and explain the earlier counter-intuitive observation that “different speaker, same word” (56.99) scored lower than “correct vs. incorrect” (60.16): a correctly-pronouncing but acoustically different speaker can score below a mispronouncing speaker whose voice happens to be closer to the reference.

V. TEAM MEMBER CONTRIBUTION

- **Mohammad Manasrah (1211407):** Speech data collection (recording protocol and all six speakers’ recordings), implementation of the preprocessing and feature extraction stage (silence removal, normalization, MFCC/pitch/formant extraction), and overall report writing and LaTeX formatting.
- **Salah Dawabsheh (1210722):** Implementation of the DTW-based reference comparison and the weighted rule-based scoring formula, design and execution of the three required experiments plus the word-mismatch control experiments, and generation of the corresponding plots and figures.
- **Osaid Nour (1210733):** Design and implementation of the machine learning classification pipeline (Logistic Regression, SVM, Random Forest), stratified cross-validation methodology, and statistical analysis of the raw-vs-similarity feature comparison and the effect-size results.

VI. CONCLUSION

We implemented a complete Arabic pronunciation assessment pipeline combining classical DSP features with a rule-based scoring formula, and validated it against three required experiments plus two additional word-mismatch control experiments and a machine learning comparison. The system reliably detects same-speaker consistency (78.6) and separates it from cross-speaker variation (57.0). The central finding of this study is a quantified limitation: speaker identity affects the score roughly twice as much as word identity (21.6 vs. 10.1 points), and removing absolute, speaker-identifying features from the ML classifiers caused accuracy to collapse from as high as 100% to as low as 58%, confirming that a substantial share of the system’s apparent discriminative power reflects speaker recognition rather than pronunciation accuracy. Future work should explore speaker-normalized features (e.g., CMVN, per-speaker pitch baselines, formant ratios instead of absolute Hz values) and a leave-one-speaker-out validation scheme with a larger and more balanced pool of speakers, to fully disentangle pronunciation accuracy from speaker identity.

AI TOOL USAGE DISCLOSURE

Claude (Anthropic) was used as a supporting assistant during this project, primarily for: (1) debugging assistance, including diagnosing a data-parsing issue caused by Windows bidirectional filename rendering; (2) explaining and clarifying

machine learning concepts (e.g., stratified cross-validation, feature leakage) used in our analysis; and (3) helping structure and format this report (IEEE template, tables, and figure layout). The system design, implementation, data collection, experiments, and final interpretation of results were carried out by the authors.

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